

Question 1– Network Scan Analysis Using Nmap

Code

```
nmap -A 192.168.1.10
```

Explanation

The -A option performs OS detection, service version detection, script scanning, and traceroute. The scan found three open ports:

- **Port 22 (SSH):** Used for secure remote login. Weak passwords or outdated SSH versions can allow unauthorized access.
- **Port 80 (HTTP):** Used to host websites. If not secured, attackers may find vulnerable web pages or exposed information.
- **Port 3306 (MySQL):** Used for database services. If publicly accessible, sensitive data may be exposed.

The operating system was detected as **Linux Ubuntu 20.04**.

Security Risks

- Weak passwords may allow unauthorized login.
- Outdated software may contain security vulnerabilities.
- Open ports increase the attack surface.
- Publicly accessible databases may expose confidential information.

Security Measures

- Use strong passwords and Multi-Factor Authentication (MFA).
- Keep the operating system and services updated.
- Enable a firewall and close unused ports.
- Restrict database access to trusted IP addresses.
- Regularly monitor system logs.

Conclusion

The Nmap scan helps identify open ports and services so administrators can secure the system before attackers exploit vulnerabilities.

Question 2– Phishing Awareness Campaign Using GoPhish

Sample Campaign

Employees Tested : 100

Clicked Phishing Link : 40

Entered Credentials : 20

Downloaded Attachment : 15

Explanation

The GoPhish report shows that many employees were unable to identify phishing emails.

- **40% clicked the phishing link**, showing a lack of awareness.
- **20% entered login credentials**, indicating they trusted a fake login page.
- **Several users downloaded a malicious attachment**, increasing the risk of malware infection.

Security Weaknesses

- Lack of phishing awareness.
- Failure to verify email senders.
- Unsafe handling of email attachments.
- Poor password security practices.

Recommendations

- Conduct regular phishing awareness training.
- Perform simulated phishing campaigns.
- Enable spam filters and email security solutions.
- Implement SPF, DKIM, and DMARC.
- Enforce strong passwords and Multi-Factor Authentication (MFA).

Conclusion

Regular employee training and strong email security controls can significantly reduce successful phishing attacks.

Question 3 (10 Marks) – DNS Assessment Using Dig

Code

```
dig AXFR example.com @ns1.example.com
```

Explanation

This command checks whether the DNS server allows a **Zone Transfer (AXFR)**. If successful, it returns all DNS records stored on the server.

Sample Output

www.example.com

mail.example.com

vpn.example.com

db01.example.com

backup.example.com

Security Implications

- Internal hostnames become visible.
- Critical servers such as mail, VPN, and database servers are exposed.
- Attackers can understand the organization's network structure.

Recommendations

- Disable unauthorized Zone Transfers.
- Allow AXFR only for trusted DNS servers.
- Use DNSSEC.
- Keep DNS servers updated.
- Monitor DNS logs regularly.
- Use firewalls to restrict DNS access.

Conclusion

A successful DNS Zone Transfer exposes important DNS records and increases security risks. Restricting AXFR requests and securing DNS servers helps protect the organization's infrastructure.